

David (Weijia) Fang

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EXPERIENCE

LOCATE ALPHA (THEN SPATIAL LASER) | ML ENGR INTERN

July – Aug 2024 | McKinney, TX

- Automatic layout and arrangement of building units based on GIS data represented in Parcel JSON.
- Arrangement of optimal traffic accessibility while maintaining existing vegetation whenever possible.

BYTEDANCE | SOFTWARE ENGR INTERN

June – Aug 2023 | Shanghai, China

- Made the usage of “Lark Toolkit” (an internal GUI toolkit easing the calling of Lark APIs) more understandable by wrapping around some API calls, such as changing a JSON-string parameter into nested JSON object.
- Split the configurations of the toolkit into a separate XLSX file as opposed to separated in different JSON documents, more user-friendly to nonprogrammers.
- Used Express and other means to create standalone scripts for frequent usage, such as migrating group chat data from Dingding (another collaboration platform) to Lark.

ANT GROUP | ALGORITHMIC ENGR INTERN

June – Aug 2021 | Shanghai, China

- Recommendation algorithm with collaborative filtering and different neural network structures.
- Natural language processing tasks: unsupervised text generation, abstraction of paper with TF-IDF and other methods, knowledge-graph-based Q&A System.

RESEARCHES

FUTURE OF PROGRAMMING LAB | RESEARCH ASSISTANT

Sept 2022 – Present | University of Michigan

- Development of small-step evaluator that shows the internal process of an interpreter more clearly.
- Improvements to case expression evaluation: debugging small-step evaluator, constraint and Incon judgments, hints for missing branches.
- Improvement in the visual cues of the user interface.
- New function syntax that separates the definition of input or output types, more closely matching other mainstream programming languages.

ALFRED HERO'S RESEARCH GROUP | GRAD STUDENT RA

Nov 2023 – Dec 2024 | University of Michigan

- Proposed a new training strategy that converges faster, reduced overfitting, which was tested on both simple Gaussian Mixture models and LLMs.
- Compared recalibration in LLM against traditional methods on tasks of binary text classification.
- Tested generalization between similar datasets

EDUCATION

UNIVERSITY OF MICHIGAN

MENG IN ELECTRICAL & COMPUTER ENGINEERING

Dec 2024 | Ann Arbor, MI

Cum. GPA: 4.0 / 4.0

UNIVERSITY OF MICHIGAN

BSC IN COMPUTER SCIENCE & PURE MATH

Apr 2023 | Ann Arbor, MI

Cum. GPA: 3.92 / 4.0

SKILLS

PROGRAMMING

6+ years:

Python • Java • C/C++

4+ years:

JavaScript • TypeScript • R

3+ years:

Rust • OCaml • Julia

TECHNOLOGY

NLP&LLM • Knowledge graphs •

Computer graphics & vision •

Neural networks • Tensorflow •

PyTorch • HuggingFace •

Git/Github • Linux

COURSEWORK

GRADUATE

Probability & Random Process

Machine Learning ECE

Image Processing

Computer Vision

UNDERGRADUATE

Intro to Operating Systems

Compiler Construction

Intro to Programming Languages

Mobile App Dev for Entrepreneurs

User Interface Development

Intro to Combinatorics

Intro to Numerical Methods

Elementary Number Theory

LINKS

Github:// [DavidFangWJ](#)

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